

Login using the default credentials (Username: admin, Password: password) and explore common vulnerabilities such as SQL injection, cross-site scripting (XSS), and file inclusion. DVWA is ideal for both beginners and experienced security professionals to hone their web application security skills as illustrated in Figure 5.

Accessing Metasploitable in command user interface:

Metasploitable is designed to simulate a vulnerable system that can be exploited using tools like Metasploit. To access the Metasploitable container, use the following command:

```
docker exec -it metasploitable /bin/sh
```

Metasploitable provides a variety of exploitable services, such as insecure FTP servers, weak database configurations, and vulnerable web services. It's an excellent environment for practising more advanced exploitation techniques.

Web access and credentials: In addition to interacting through the terminal, some of these services can be

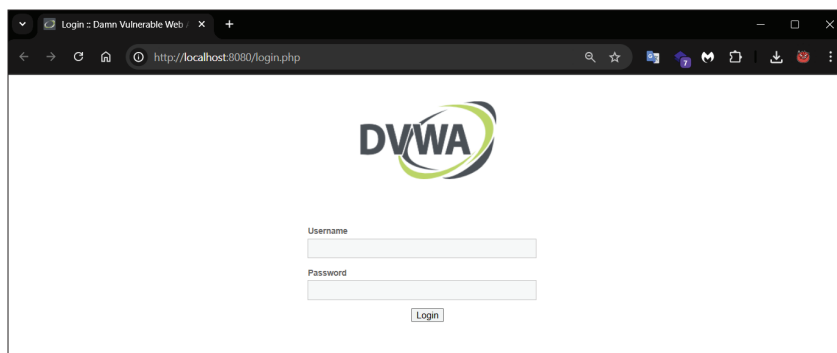


Figure 4: DVWA in Windows browser

accessed via a web browser.

- DVWA: `http://localhost:7001`
- DVWA: Username: admin, Password: password

By following these steps, you now have a fully functional security lab where you can test, practice, and explore various security tools and techniques. Docker Compose ensures that this environment is easy to manage and can be brought up or taken down with minimal effort.

Now, as we have started all the containers, Kali is running with the IP 172.19.8.5; DVWA is running with the IP 172.19.8.6; and Metasploitable is

running with the IP 172.19.8.4. After starting the containers, the devices can successfully ping with each other within the Docker network. This confirms that all the machines are interconnected and can communicate securely inside the isolated testbed environment as illustrated in Figure 5.

Let us check the basic connectivity between the Docker images. Figure 6 clearly shows that attempting to ping the container IPs (such as 172.19.8.5 and 172.19.8.6) from the Windows host results in 'Request timed out' and 100% packet loss; however, the containers themselves can successfully ping

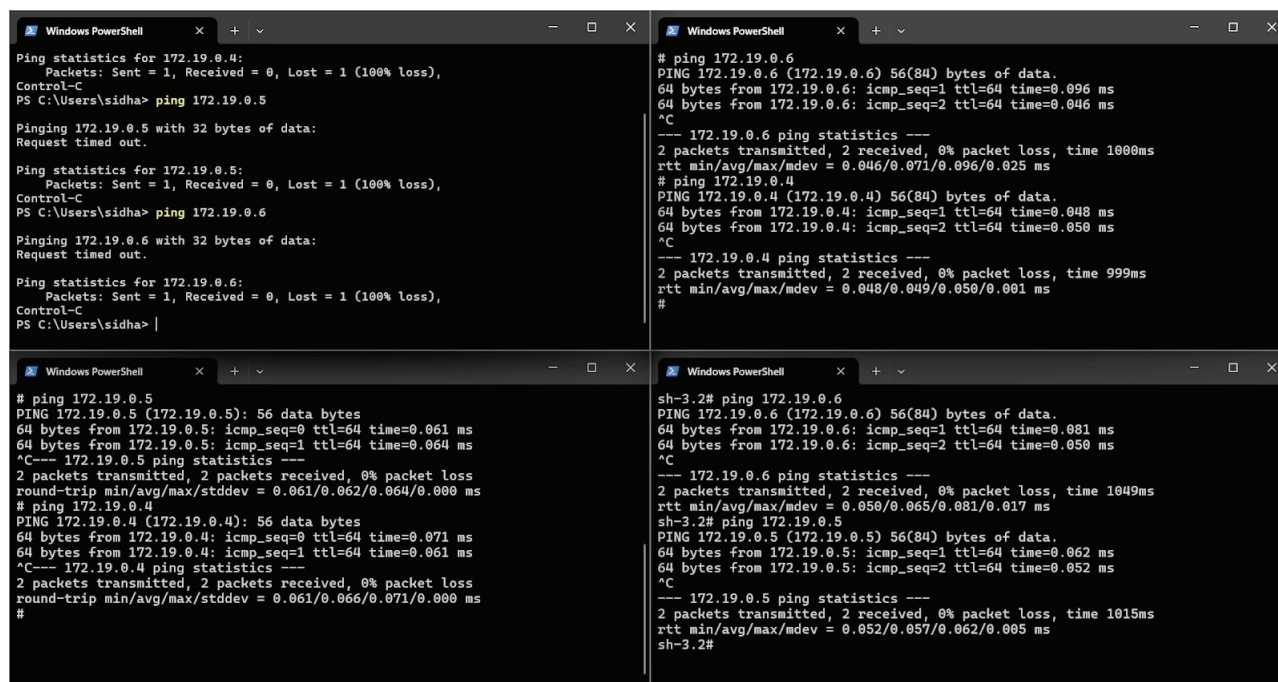


Figure 5: Security testbed with three containers and one host machine